Name: ______________________

DECIMALS, (MULTIPLYING DECIMALS)

Date: ________________

LO: I can multiply decimal numbers using place value and the column method.

Multiplying decimals

To **multiply decimals**, ignore the decimal points, multiply as whole numbers, then put the point back using the total number of **decimal places**.

Rule: Count the total decimal places in the question. The answer has the same total number of decimal places.

Key method

●	0.3 × 0.4 = 0.12	1 d.p. + 1 d.p. = 2 d.p.
●	2.5 × 4 = 10.0	1 d.p. + 0 d.p. = 1 d.p.
●	1.2 × 0.3 = 0.36	12 × 3 = 36, 2 d.p.
●	0.3 × 0.4 = 1.2	wrong number of d.p.
●	2.5 × 4 = 100	forgot the decimal point

Key idea

Multiply as whole numbers, then place the decimal point so the answer has the correct total number of decimal places.

1.2 × 0.3: 12 × 3 = 36 0.36

2 decimal places in total

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – decimal by whole number

Calculate 3.4 × 5

Ignore decimal: 34 × 5 = 170

Total d.p. in question: 1

Place point: 17.0

= 17.0

One decimal place in the question means one in the answer.

Example 2 – decimal by decimal

Calculate 0.6 × 0.7

Ignore decimals: 6 × 7 = 42

Total d.p.: 1 + 1 = 2

Place point: 0.42

= 0.42

Two decimal places total: answer is 0.42 not 4.2.

Example 3 – two-digit decimals

Calculate 2.4 × 1.3

Ignore decimals: 24 × 13 = 312

Total d.p.: 1 + 1 = 2

Place point: 3.12

= 3.12

24 × 13: 24×10 = 240, 24×3 = 72, total 312.

Example 4 – with zeros

Calculate 0.05 × 0.4

Ignore decimals: 5 × 4 = 20

Total d.p.: 2 + 1 = 3

Place point: 0.020 = 0.02

= 0.02

Three decimal places needed; 020 becomes 0.020 = 0.02.

YOUR TURN – PRACTICE (deliberate practice)

1. Decimal × whole number
Calculate.

a) 2.3 × 4

b) 5.6 × 3

c) 0.7 × 8

d) 3.25 × 2

2. Decimal × decimal (1 d.p. each)
Count decimal places carefully.

a) 0.3 × 0.4

b) 0.6 × 0.5

c) 1.2 × 0.3

d) 2.5 × 0.4

3. Harder multiplications
Work as whole numbers first, then place the point.

a) 2.4 × 1.3

b) 3.5 × 2.6

c) 4.2 × 0.15

d) 0.08 × 0.6

4. Word problems
Multiply to find the answer.

a) Petrol costs £1.45/litre. Cost of 8 litres?

b) A tile is 0.3 m × 0.3 m. What is its area?

c) Speed 4.5 km/h for 2.5 hours. Distance?

d) 3 items at £2.99 each. Total?

5. Mixed multiplication
Calculate each product.

a) 6.5 × 4 **b) 0.4 × 0.4** **c) 1.5 × 1.5** **d) 0.12 × 0.3** **e) 7.2 × 0.5**

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) 0.3 × 0.4 = 0.12 (2 d.p.) ☐ Correct answer: __________ Reason: __________________	**b) 0.3 × 0.4 = 1.2** ☐ Correct answer: __________ Reason: __________________	**c) Total d.p. in answer = sum of d.p. in question** ☐ Correct answer: __________ Reason: __________________	**d) Multiplying decimals always gives a bigger number** ☐ Correct answer: __________ Reason: __________________	**e) 2.5 × 4 = 10** ☐ Correct answer: __________ Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A square has side length **3.5 cm**. Find its area. Then find the area of a square with side **0.35 cm**. How many times smaller is the second area?

Expression: ______________________________

Simplified: ______________________________

b) Without a calculator, work out **2.4 × 3.5** and **0.24 × 35**. Explain why the answers are the same. What about **24 × 0.35**?

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!